

Pravin Salla

Davis, CA | +1(925)209-8261 | prsalla19@gmail.com | [linkedin.com/in/pravin-salla-312609264](https://www.linkedin.com/in/pravin-salla-312609264) | Portfolio: pravinsalla.dev

EDUCATION

University of California, Davis - M.S. Mechanical & Aerospace Engineering (GPA: 3.7/4.0)

Sept 2024 - Dec 2026

Relevant Coursework: Advanced Manufacturing, Advanced Mechanical Design, Mechanical Performance of Materials, Modern Manufacturing Techniques

Somaiya Vidyavihar University - B.Tech Mechanical Engineering, Honors in Design (GPA: 3.5/4.0)

Sept 2020 - June 2024

Relevant Coursework: Mechanical System Design, Process Equipment Design, Failure Analysis, Automation, Elements of Machine Design, Strength of Materials, Manufacturing of Composites

TECHNICAL SKILLS

Programming/ Data: MATLAB, Simulink, Python, Arduino IDE, MS Excel

Engineering Methods: FEA, FMEA, GD&T, Root Cause Analysis, Tolerance Analysis, DFM, CFD, Failure Analysis, Design Validation, Technical Documentation, DOE, Product Life Cycle Assessment

CAD / Simulation: SolidWorks, AutoCAD, Fusion 360, ANSYS, CATIA, PTC Creo, Siemens NX, Abaqus

Testing / Controls: LabVIEW, DAQ, Sensor Integration, Signal Processing, NI Signal Express, Frequency-Domain Analysis, Arduino, Raspberry Pi

Manufacturing / Hardware: CNC Machining, 3D Printing (FDM, SLA, SLS), Laser Cutting, Milling, Lathe, Grinding, Welding, Injection Molding, Compression Molding, Soldering, Pneumatics & Hydraulics

WORK EXPERIENCE

Graduate Student Researcher - AHMCT Research Center, UC Davis (in collaboration with Caltrans)

Apr 2026 - Present

Sacramento, California

- Conduct applied research on heavy equipment safety systems for industrial and construction applications with Caltrans.
- Support design, integration, and validation of remote shutdown systems for Volvo and CAT wheel Loaders.
- Perform instrumentation setup, pilot testing, and data acquisition for proposed safety systems.
- Contribute to technical reports, engineering documentation, and recommendations for field deployment.

Graduate Teaching Assistant - EME 50: Manufacturing Processes, UC Davis

Aug 2025 - Mar 2026

Davis, California

- Conducted hands-on instruction in manual milling, lathe, drill press and CNC machining, covering setup, tooling, machining parameters, and safe shop practices for the EME-50 course for 3 quarters.
- Instructed CAD modelling and CAM programming in Fusion 360, reviewing part models and troubleshooting toolpaths and work-holding setups, as well as reviewing technical drawings and application of GD&T principles.
- Led machining demonstrations for multi-component gyroscope assemblies, reinforcing machining fundamentals and process consistency.
- Guided students through part inspection workflows, including dimensional verification, completion of inspection sheets, and pass/fail quality assessments against drawing tolerances

Design Intern - Ayka Control Systems

June 2023 - April 2024

Mumbai, India

- Contributed to prototyping and design iterations for an item storage and tracking system
- Reviewed lead designer's part drawings for GD&T compliance and manufacturability, reviewing 40+ drawings, flagging issues before release.
- Translated client design concepts into 3D CAD models across a range of components including housings and mechanical parts, producing 3D-printed prototypes for client review and approval
- Collaborated with the electronics team to design enclosures and internal layouts for EV chargers, routing components and wiring to meet spatial and assembly requirements

ACADEMIC PROJECTS

Hahn Grinder Friction Characterization & Data Analysis - MASTeR Lab, UC Davis

Jan 2025 - June 2025

- Characterized friction on a Cylindrical Grinder linear axis using a reciprocating sled setup, integrating load cell and potentiometer signals via NI DAQ across 8 piston-speed levels under lubricated and non-lubricated conditions.
- Developed MATLAB scripts to convert raw voltage to physical units, segment strokes, and analyze friction - velocity trends and cycle-to-cycle repeatability, producing structured datasets for servo sizing and tribological studies.

Subsonic Wind Tunnel

Jan 2024 - April 2024

- Co-designed and fabricated a low-speed, suction-type wind tunnel from first principles (contraction cone, honeycomb settling chamber, acrylic test section and diffuser) in concept (SOLIDWORKS) before manufacturing and assembling the physical system.
- Validated the design in ANSYS Fluent (k-omega, ~836K-node mesh), achieving 9.4 m/s test-section velocity at turbulence intensity as low as 0.42%, and confirmed results via smoke visualization and anemometer testing on 3D-printed models.

Rotating Machinery Fault Diagnosis & Vibration Analysis - Published in Noise & Vibration Worldwide (SAGE Pub)

Sep 2023 - Dec 2023

- Conducted fault-diagnosis testing on a Machinery Fault Simulator (1.4 - 4.53 g unbalances, 500 - 1500 RPM, single and multi-plane) and performed FFT-based vibration analysis to quantify effects of unbalance mass, speed, and angular separation.
- Built SolidWorks rotor models and ANSYS FE/modal analysis validating natural frequencies within 10% of experimental results; co-authored the resulting peer-reviewed article.